Midterm Report: Social Context and Judgment Revision in Language Models

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Abstract

When AI agents review each other's code or compare research findings, they must evaluate conflicting judgments. Reliable collaboration requires correcting errors without abandoning correct answers in response to misleading arguments. We study how social context affects judgment revision in large language models (LLMs), varying whether prior answers are described as publicly shared and whether peer counterarguments use neutral or controlling language.

Using counterfactual replay, we hold questions and counterarguments fixed while varying contextual framing. Pilots with Llama 3.1 8B, Gemma 3 12B, and Qwen3.5 9B show that controlling language increases peer-answer adoption in Llama and Qwen, while public disclosure has little overall effect. The models differ in their responses to source attribution and their susceptibility to harmful revisions; Gemma's limited sample does not support a detailed comparison. In re-ask controls, telling Llama its prior answer was shared increases spontaneous answer changes relative to describing it as recorded.

Follow-up experiments will test whether attribution and disclosure effects persist across source identities and alternative prompt formulations.

Keywords AI alignment · multi-agent systems · judgment revision · social influence · language models

1 Introduction

In a team of AI coding agents, one agent might implement a function while another reviews it and proposes a correction. In a research workflow, agents might consult different sources and reach conflicting conclusions. In both cases, an agent must assess whether another agent's reasoning warrants revising its own judgment. Correcting an error can improve the team's output, but deferring to a forceful yet misleading argument can introduce new errors.

We examine whether social cues affect this revision process. Our experiments vary two factors: whether a model's initial answer is described as private or publicly shared, and whether a conflicting argument uses neutral or controlling language. The design draws on research on social influence and psychological reactance [1, 2], without assuming that similar behavior in LLMs reflects the same psychological mechanisms. The question also relates to sycophancy: agreement with another party can come at the expense of accuracy. Here, we study responses to peer agents rather than agreement with a user's stated beliefs.

Prior work on LLM conformity and multi-agent debate distinguishes spontaneous answer changes from peer influence and reasoning-based persuasion [3, 4]. Source identity and anonymization can also affect judgment revision [5], and persuasive strategies can influence model behavior [6, 7]. It

remains unclear how directive language interacts with source attribution and the public disclosure of an earlier answer.

We use controlled counterfactual replay on multi-hop question-answering tasks to compare responses to the same counterargument under different social cues. We distinguish beneficial corrections from harmful revisions to assess when these cues improve accuracy and when they lead models to abandon correct answers.

2 Methodology

2.1 Experimental Framework

We use counterfactual replay to compare how LLMs respond to conflicting answers under controlled conversational conditions. Independent agents first answer the same question without access to one another's responses. We identify *disagreement events*, where a recipient model and a peer produce different answers, and replay each event across conditions. The question and substantive counterargument remain fixed; only the contextual framing changes.

The primary design crosses prior-answer disclosure (private versus public) with counterargument framing (neutral versus controlling). In public conditions, the recipient is told that its previous answer was shared with other agents. Controlling counterarguments direct the recipient to accept the alternative answer, while neutral counterarguments present the disagreement without that pressure.

Following work separating spontaneous answer changes from peer-induced revision [3, 4], we include controls that remove speaker attribution, provide only the peer's proposed answer, or ask the recipient to reconsider without a counterargument. A recorded-answer condition separates acknowledgment of a prior response from its stated public disclosure. We also test an independence-oriented persona to examine whether emphasizing independent judgment changes responses to controlling language.

2.2 Datasets and Models

Our initial pilot used ARC-Challenge [8] with Llama 3.1 8B Instruct. We then moved to HotpotQA [9], where questions require integrating information across multiple supporting passages. These tasks allow us to examine disagreements involving multi-step reasoning.

The latest pilots evaluate Llama 3.1 8B, Gemma 3 12B, and Qwen3.5 9B on a common pool of 1,200 questions balanced across six question categories. Each model generates its own disagreement events, which we replay under 18 conditions. After screening and exclusions, the main comparisons include 131 events for Llama, 18 for Gemma, and 84 for Qwen. Before final exclusions, the runs produced 24,300, 3,240, and 15,120 replay responses, respectively.

2.3 Evaluation

Our primary outcome is *answer adoption*: whether the recipient replaces its initial answer with the peer's proposed answer. We estimate framing and disclosure effects, including their interaction, through event-level comparisons with cluster bootstrap confidence intervals.

We separately measure *beneficial corrections*, where an incorrect answer is replaced by a correct one, and *harmful revisions*, where a correct answer is replaced by an incorrect alternative. Re-ask controls measure answer instability in the absence of an external counterargument.

3 Preliminary Results

We initially expected controlling language to produce resistance to a peer's answer. Instead, it increased adoption in both Llama and Qwen. Public disclosure had little overall effect on adoption across the three models, providing no clear evidence that publicly sharing an answer makes a model more resistant to disagreement. Gemma's estimated framing effect was small (+1.7 percentage points), but its 18 usable events are insufficient for a detailed behavioral comparison. Table 1 summarizes Llama and Qwen.

Table 1: Pilot behavior in Llama and Qwen. Adoption effects are measured in percentage points (pp); harmful revision rates are conditional on the initial answer being correct. Ranges summarize the reported rates across disclosure conditions. Each model generated its own disagreement events, so these are descriptive comparisons rather than estimates on a matched event set.

Measure	Llama 3.1 8B	Qwen3.5 9B
Usable disagreement events	131	84
Adoption increase under controlling language	+12.8 pp	+16.7 pp
95% confidence interval	[9.1, 16.4] pp	[11.7, 22.4] pp
Harmful revisions, neutral framing	71–72%	19–20%
Harmful revisions, controlling framing	86–89%	39–40%
Effect of naming a peer on the controlling-language effect	Amplifies by 5.3 pp	Reduces by 5.3 pp
Re-ask answer changes: recorded versus shared	15.5% versus 27.6%	Little difference observed

3.1 Adoption and Accuracy

The adoption increase includes harmful revisions, not just successful corrections. Under neutral framing, Qwen corrected approximately 85–89% of initially incorrect answers while abandoning 19–20% of initially correct answers. Controlling language raised its harmful revision rate to 39–40%. Qwen was more stable than Llama in retaining initially correct answers under both neutral and controlling framing. Because each model generated its own disagreement events, this comparison is descriptive rather than evidence of an intrinsic difference in stability. Within each model's event set, directive wording therefore increased susceptibility to misleading counterarguments.

3.2 Attribution and Baseline Stability

Naming a peer changed the controlling-language effect in opposite directions: it amplified the effect in Llama and reduced it in Qwen, relative to presenting the argument without an attributed speaker. These estimates suggest that source attribution may alter how models respond to directive wording.

Disclosure produced a separate pattern in Llama's re-ask controls. Without a counterargument, describing its previous answer as shared increased answer changes from 15.5% to 27.6%, compared with describing it as recorded. Qwen showed little difference. This result leaves open whether disclosure affects baseline answer stability even when it has little effect on peer-answer adoption.

4 Challenges & Open Questions

The main interpretive challenge is separating responses to social context from ordinary instruction-following. Controlling language explicitly encourages adoption, so its effect may reflect compliance with a directive rather than social pressure. Speaker-free and stance-only controls help assess these explanations. Likewise, the recorded-versus-shared difference may depend on particular wording rather than public disclosure itself.

The model comparisons also have limits. Each model generates its own disagreement events, so the selected questions and counterarguments differ. We cannot attribute the observed differences solely to model behavior without a matched set of events. Gemma's sample is especially small, partly because of difficulties obtaining responses in the required format.

Answer correctness is also an incomplete measure of revision quality. A correct final answer does not establish that the model evaluated the counterargument appropriately. Because the pilots include many exploratory comparisons, subgroup patterns need validation in follow-up experiments.

Finally, a behavioral resemblance to human social influence does not establish a shared psychological mechanism [10]. Our follow-ups will test whether the observed effects persist when we vary the wording and source of an argument while preserving its substantive content.

5 Next Steps

We will prioritize two follow-up experiments based on the attribution and stability differences in Table 1. Both will retain answer adoption, beneficial correction, and harmful revision as outcomes.

5.1 Controlling Language and Source Attribution

The opposite attribution patterns in Llama and Qwen motivate a closer test of how source identity interacts with directive language. Prior work shows that agent identity can affect revision [5], while substantial updating can occur without an attributed speaker [4]. We will extend the factorial comparison of neutral versus controlling language and named-peer versus speaker-free arguments, keeping the proposed answer and reasoning fixed.

We also plan to compare anonymous sources and domain experts to test whether apparent expertise changes the framing effect. The central question is whether directive wording has a consistent effect across sources or whether its influence depends on the authority attributed to the speaker.

5.2 Recorded Versus Shared Answers

Llama's higher re-ask instability under shared-answer wording motivates a separate disclosure experiment. We will compare a plain re-ask baseline with private, recorded, and shared conditions, using multiple formulations of each statement. This will test whether the difference persists across wording changes or depends on specific terms.

We will then introduce neutral and controlling counterarguments while holding their reasoning fixed, to assess whether disclosure also affects responses to external disagreement. Work on social context and persuasive framing informs this design [1, 6], but the immediate goal is to establish how robust the pilot effect is.

If these effects replicate, we will test additional model families and tasks beyond HotpotQA. We also plan to evaluate Inspect AI as an evaluation framework and compare judging approaches for

open-ended responses. We will prioritize replication and prompt variation before expanding to a broader set of tasks.

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